Ex.no -3.2- CREATE 3D MODEL FROM ORTHOGRAPHIC PROJECTION

AIM:

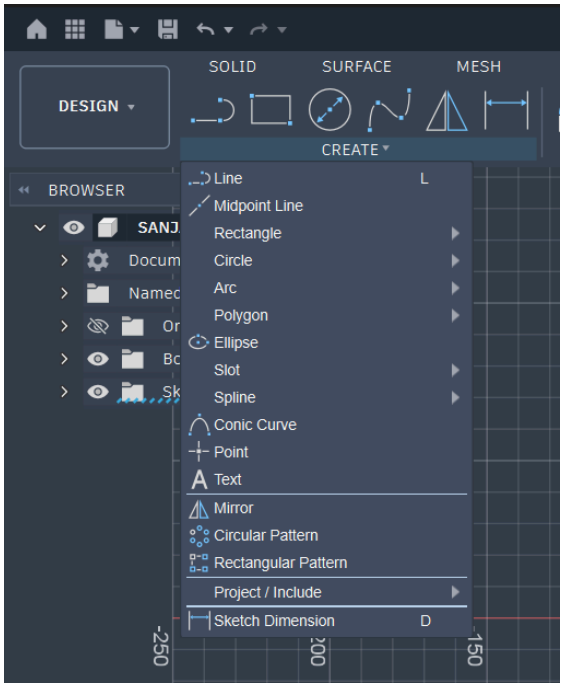
To sketch the given drawing with dimensions using fusion 360 tool.

REQUIREMENTS:

1. Autodesk fusion 360
2. Windows 11
3. 16 GB of RAM (integrated graphics recommend 2 GB or more)
4. 2.5 Mbps or faster download; 500 Kbps or faster upload

PROCEDURE:

1. Open a new file and then select the sketch icon.
2. Select the plane for sketching.
3. Use sketching features (line, circle, arc, rectangle, polygon, etc.,) to draw the given sketch.



Top View (Plan View)

- **300 mm:** Three hundred millimeters (distance between the centers of the two end holes)
- **150 mm:** One hundred fifty millimeters (half-distance from the center line to the right end hole)
  40 mm: Diameter forty millimeters (outer circular end tabs)
- **20 mm:** Diameter twenty millimeters (center hole)
- **150 mm:** One hundred fifty millimeters (total width of the main central body section)
- **R 60 mm:** Radius sixty millimeters (rounded ends on the left and right mounting tabs)

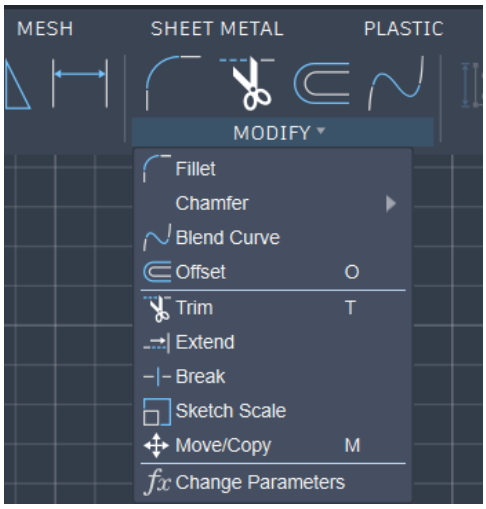
Front Elevation View

- **420 mm:** Four hundred twenty millimeters (total overall length of the base flange)
- **R 110 mm:** Radius one hundred ten millimeters (outer curved arch)
- **120 mm:** One hundred twenty millimeters (width of the inner rectangular cutout at the bottom)
- **30 mm:** Thirty millimeters (thickness of the base flange)

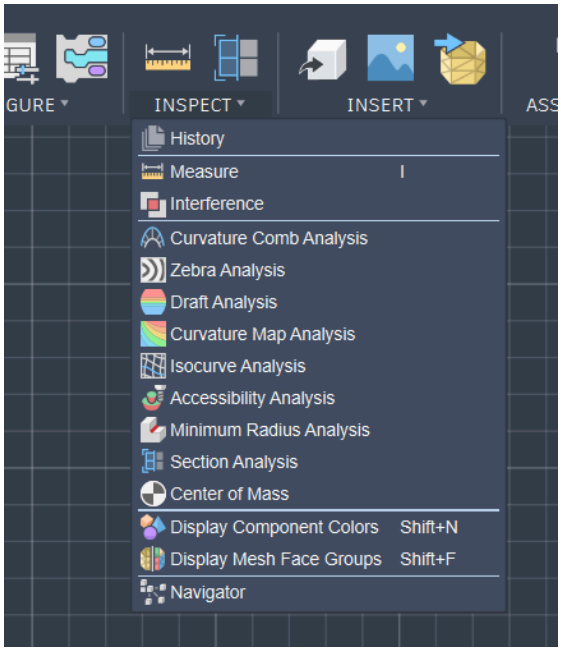
Side View (End Elevation)

- **150 mm:** One hundred fifty millimeters (total width/depth of the side profile)
- **110 mm:** One hundred ten millimeters (total height from the bottom base to the top)
- **60 mm:** Sixty millimeters (height from the bottom base to the center horizontal axis/dashed line)

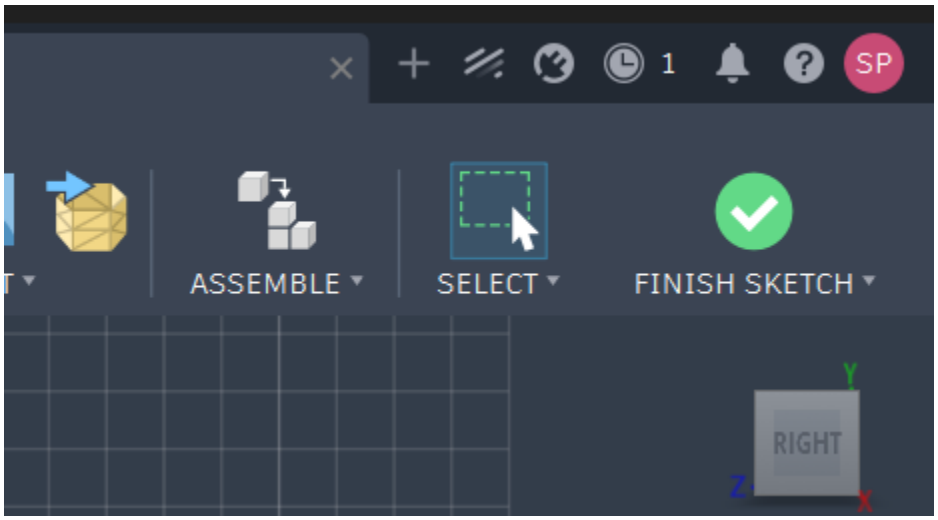
4. Use sketch modifying features (cut, trim, mirror, move, etc.,) to edit the sketch.



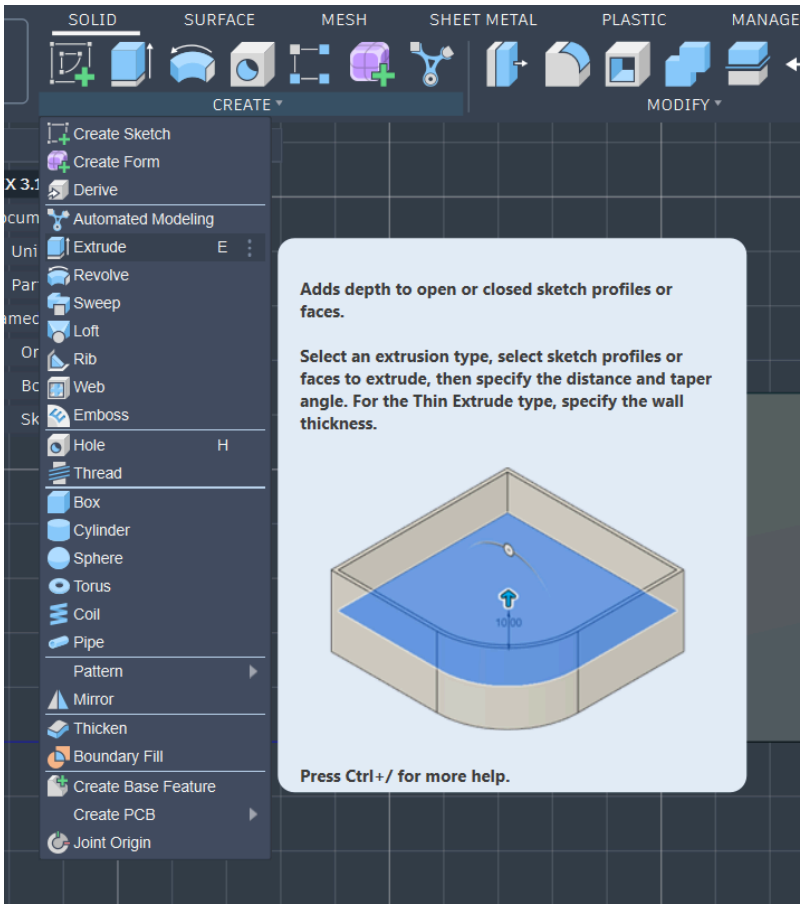
5. Use inspect feature to measure the dimensions of the sketch.



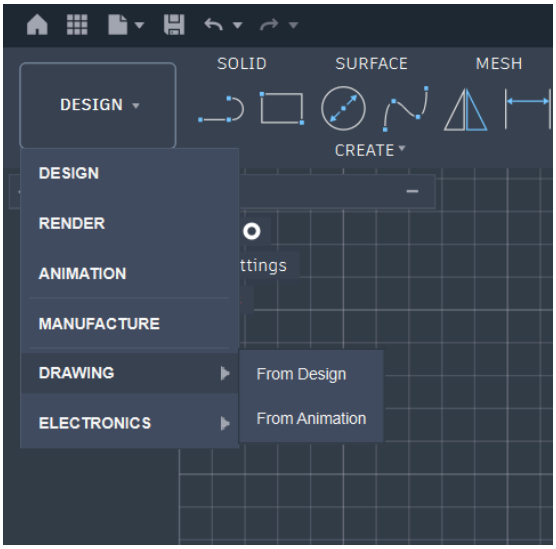
6. After finishing the sketch use the finish sketch option to finalize the sketch.



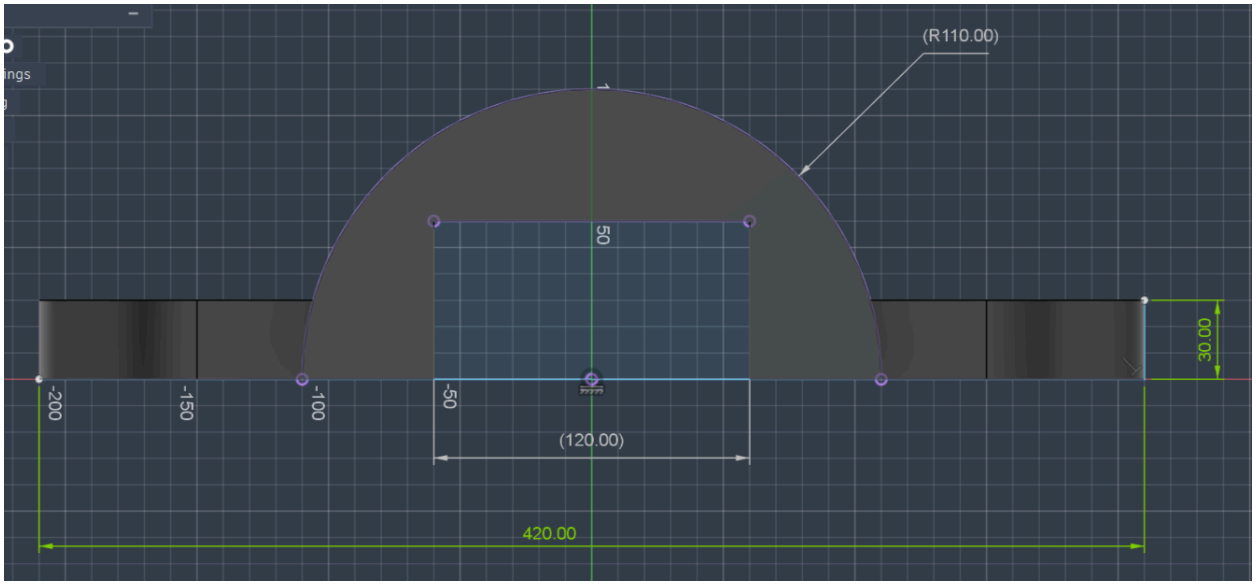
7. Use extrude option to convert 3D model from sketch

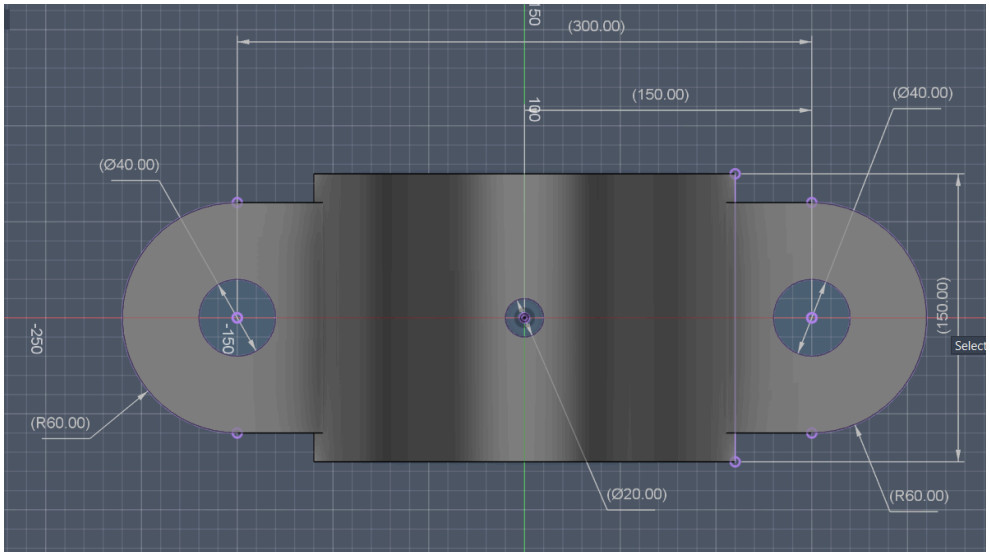


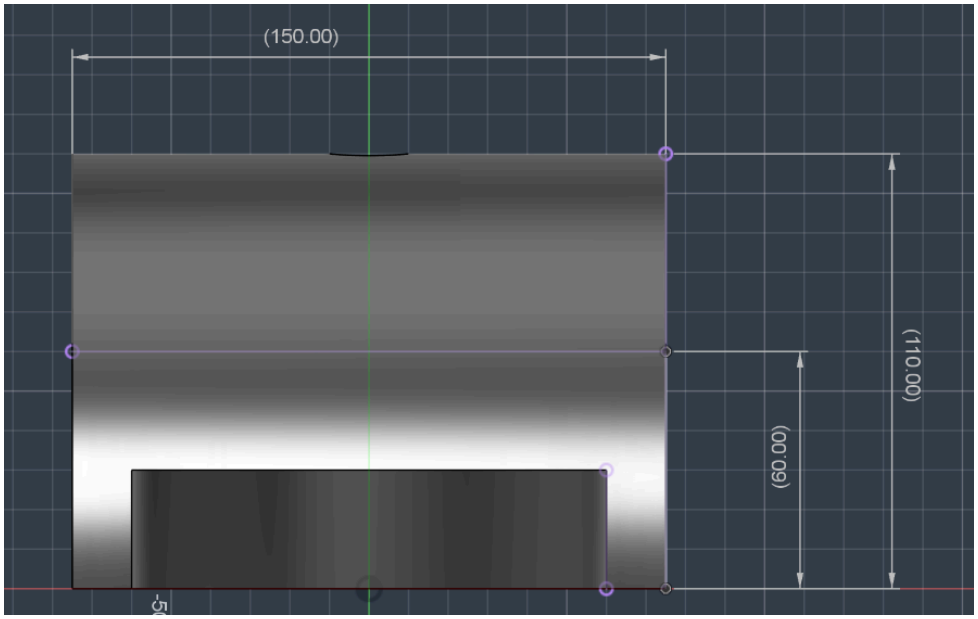
8. After finishing the model convert to drawing sheet



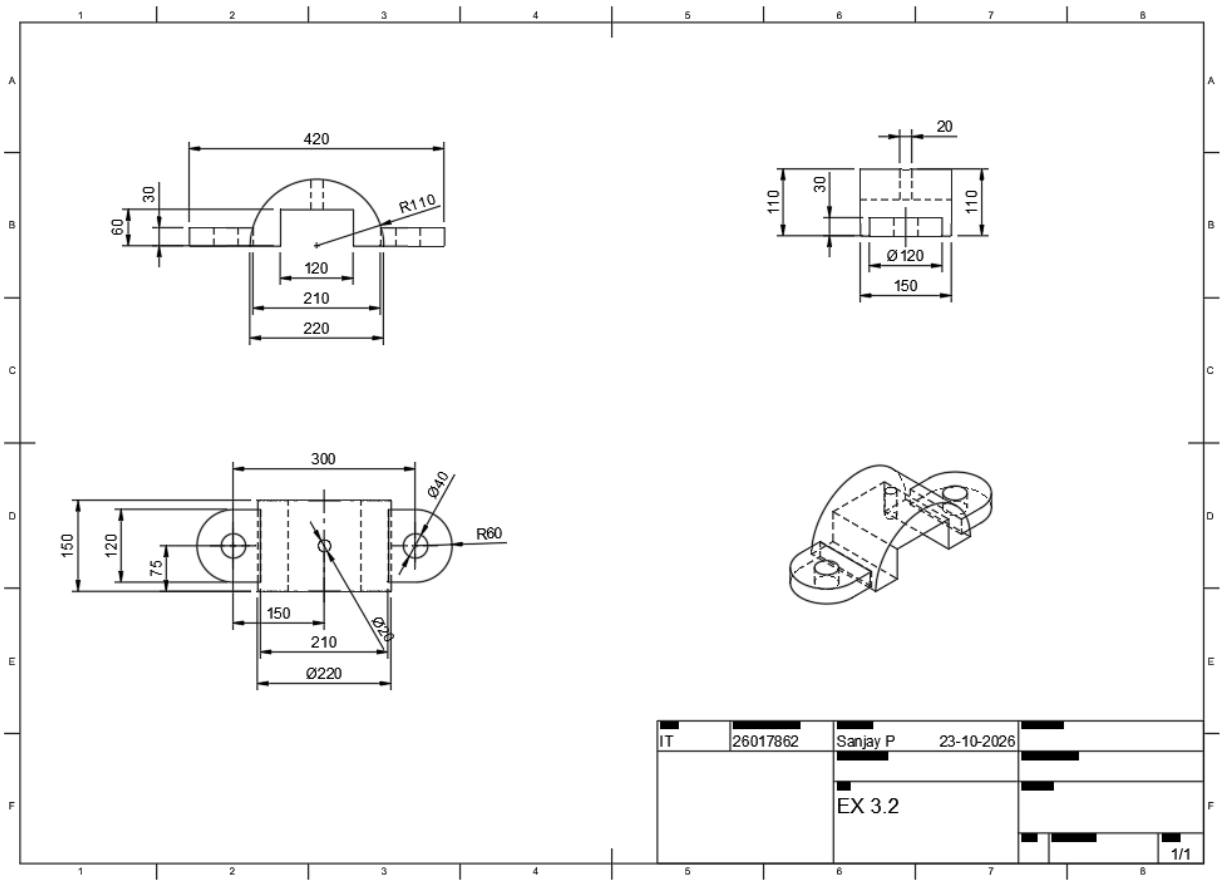
INPUT:




150
(300.00)
100
(150.00)
(Ø40.00)
(Ø40.00)
-250
-150
(150.00)
(R60.00)
(Ø20.00)
(R60.00)


(150.00)
(110.00)
(60.00)

OUTPUT:



RESULT:

Thus, the 3D model is created using orthographic projection.